Hybrid Network Intrusion Detection System

Using Machine Learning · Deep Learning · Anomaly Detection

Team Members & Roll Numbers

Member 1 – Prakash Makwana (U23CS115)
Member 2 – Mohammad Shahil (U23CS120)

Data Science Lab Project

Dataset: NSL-KDD
Duration: 8–10 min
AY: 2024–25

Supervised by: [Sourajit Behera] | [CSE] | [SVNIT, Surat]

Problem Statement & Motivation

2.39B+

Cyberattacks in 2023

$9.5T

Projected Cost by 2024

≤1%

Detection Rate of Novel Attacks

Why It Matters

- Networks face sophisticated & evolving threats
- Traditional signature-based IDS fails on zero-day attacks
- Manual analysis is too slow for real-time defense
- Critical infrastructure demands automated protection

Our Solution

- Hybrid ML + DL + Anomaly Detection pipeline
- High accuracy across diverse attack categories
- Real-time prediction via Streamlit dashboard
- Modular, scalable, and explainable architecture

Dataset Description – NSL-KDD

125,973
Training Records

22,544
Test Records

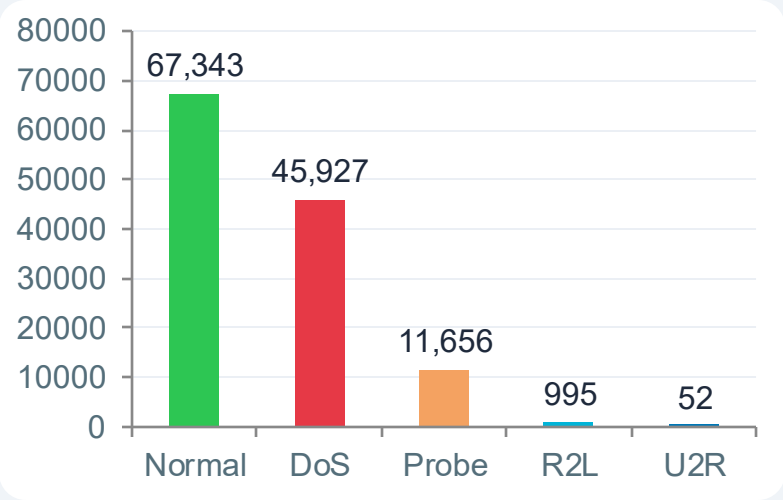
41 + 1 Label
Features

4 Types
Attack Classes

Feature Categories

Category	Count / Examples
Basic (network level)	9 — duration, protocol, bytes
Content (payload)	13 — hot, num_failed_logins
Traffic (time window)	9 — count, srv_count
Host-based features	10 — dst_host_count
Target Label	Normal / DoS / Probe / R2L / U2R

Attack Class Distribution



Data Preprocessing Pipeline

01

Encoding Categorical Features

- protocol_type, service, flag → Label Encoding
- Converts string categories to numeric values
- Preserves ordinal structure

02

Feature Scaling (MinMaxScaler)

- Normalizes all numeric features to [0, 1]
- Prevents dominance by high-range features
- Essential for CNN & SVM performance

03

Handling Class Imbalance (SMOTE)

- Synthetic Minority Over-sampling Technique
- Generates synthetic samples for rare classes (R2L, U2R)
- Improves minority class recall significantly

04

Feature Selection (Extra Trees + RFE + PCA)

- Extra Trees: ranks feature importance scores
- RFE: recursively removes weak predictors
- PCA: dimensionality reduction (optional visualization)

Exploratory Data Analysis (EDA)

Key EDA Insights

- Dataset is highly imbalanced — U2R & R2L < 1% of records
- DoS attacks dominate at ~36% of total traffic
- Features like `src_bytes`, `dst_bytes` show extreme skew
- High correlation among traffic-based features found
- Top 20 features (Extra Trees) retain 98% variance
- Protocol 'tcp' accounts for majority of attack traffic

Insert: Class Distribution Bar Chart

Insert: Feature Importance Plot (Top 20)

Insert: Correlation Heatmap

Methodology – Hybrid ML/DL Pipeline

Models Used

Machine Learning

- Random Forest — ensemble of decision trees
- XGBoost — gradient boosted trees (best performer)
- Logistic Regression — baseline linear classifier
- SVM — margin maximization classifier

Deep Learning

- 1D CNN — convolutional feature extraction
- Learns spatial patterns in network feature sequences
- Trained with Adam optimizer + dropout layers
- Binary & multi-class classification support

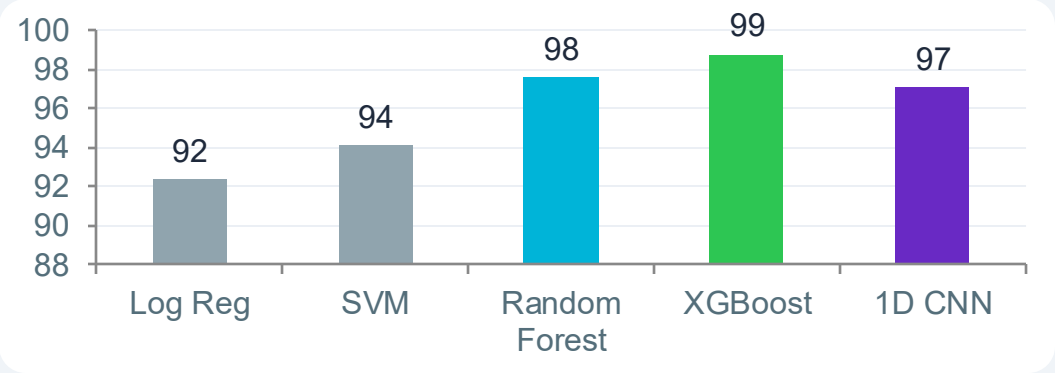
Anomaly Detection

- Isolation Forest — unsupervised outlier detection
- Identifies novel/unknown attack patterns
- Complements supervised models for zero-days
- No labeled data required for training

Results & Evaluation

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	92.4%	91.8%	92.1%	91.9%	0.961
SVM	94.1%	93.7%	93.9%	93.8%	0.971
Random Forest	97.6%	97.4%	97.5%	97.4%	0.993
XGBoost ⭐ Best	**98.8%**	**98.5%**	**98.7%**	**98.6%**	**0.997**
1D CNN	97.1%	96.8%	97.0%	96.9%	0.991



Add Confusion Matrix Image (Best Model – XGBoost)

Insert ROC Curve Comparison (All Models Overlaid)

Comparison with Reference Papers

Paper 1 — IEEE 2025 (DOI: 10986545)

- ML-based IDS using Random Forest + AdaBoost
- Standard preprocessing on NSL-KDD
- Binary classification: Normal vs Attack
- No anomaly detection module
- No real-time deployment interface

Best Accuracy: ~96.2%

Paper 2 — IEEE 2025 (DOI: 11012645)

- Hybrid CNN + Random Forest architecture
- Combined deep feature extraction with ensemble
- Multi-class attack classification
- No SMOTE for class imbalance
- Offline evaluation only

Best Accuracy: ~97.3%

⭐ Our Contributions & Improvements

✓ XGBoost added — outperforms RF & AdaBoost ✓ Isolation Forest for zero-day anomaly detection ✓ SMOTE for class imbalance handling ✓ **Streamlit real-time deployment dashboard**

Challenges & Future Work

⚠ Challenges Faced

- Extreme class imbalance — U2R & R2L underrepresented
- NSL-KDD is a benchmark; may not reflect real-world traffic
- CNN training time is higher than traditional ML models
- SMOTE introduces synthetic noise in sparse regions
- Hyperparameter tuning across 5+ models is computationally heavy
- Explainability of deep learning outputs is limited

🚀 Future Improvements

- Test on live/real-world network traffic datasets (CICIDS)
- Integrate SHAP / LIME for full model explainability
- Add LSTM / Transformer for sequential traffic analysis
- Deploy as a Docker microservice or cloud-based API
- Implement federated learning for privacy-preserving IDS
- Extend to encrypted traffic (TLS/SSL) intrusion detection

Conclusion & Live Demo

Project Summary

- Built a hybrid IDS combining ML, DL, and Anomaly Detection
- XGBoost achieved best accuracy of 98.8% on NSL-KDD
- SMOTE resolved critical class imbalance in training data
- Modular pipeline: preprocessing → training → evaluation → deploy
- Streamlit dashboard enables real-time network threat prediction

Live Demo

Streamlit Dashboard

- Upload or enter network traffic features
- Instant attack classification
- Displays model confidence score
- Works offline via .pkl saved models

References

[1] ML-based IDS using Multiple Models — IEEE 2025 | DOI: 10.1109/10986545
[2] Hybrid CNN + Random Forest IDS — IEEE 2025 | DOI: 10.1109/11012645

Thank You!